

matrixdiagrams.sty

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Overview

`matrixdiagrams` draws schematic matrix diagrams (bracketed boxes with diagonals, shaded blocks, labels, and partition lines) directly in TikZ, for typesetting matrix decompositions. Every diagram is built with one command, `\matrixdiagram[scale][aspect]{content}`, where `content` is a sequence of small drawing commands (`\mdFill`, `\mdLabel`, `\mdDiagBand`, ...) working in a normalized coordinate system that automatically adapts to the diagram's scale and aspect ratio. The sections below give a quick reference followed by every command with a runnable example.

1 Quick Reference

<code>\matrixdiagram [s][a]{c}</code>	Main command. <code>s</code> = scale (default 1), <code>a</code> = aspect (default 1). Position args below are fractions $x, y \in [0, 1]$, any aspect.
<code>\mdDiag</code>	Main diagonal
<code>\mdDiagBand</code>	Tridiagonal band.
<code>\mdDiagBandN {n}</code>	n -diagonal band (n odd).
<code>\mdBidiagUpper</code>	Main diagonal + super-diagonal.
<code>\mdBidiagLower</code>	Main diagonal + sub-diagonal.
<code>\mdHessenberg</code>	Upper-triangular outline + sub-diagonal.
<code>\mdHessenberg [color]</code>	Same with filled interior.
<code>\mdDiagFrom {x1}{y1}{x2}{y2}</code>	Main diagonal from (x_1, y_1) to (x_2, y_2)
<code>\mdDiagBandFrom {x1}{y1}{x2}{y2}</code>	Tridiagonal band, clipped.
<code>\mdDiagBandNFrom {n}{x1}{y1}{x2}{y2}</code>	n -diagonal band, clipped.
<code>\mdFill [col]{x0}{y0}{x1}{y1}</code>	Filled rectangle. Default color <code>gray!20</code> .
<code>\mdVLine {x}</code>	Vertical line, full height.
<code>\mdHLine {y}</code>	Horizontal line, full width, at fractional height y .
<code>\mdLabel {x}{y}{text}</code>	Math-mode label at (x, y) .
<code>\mdLabel [color]{x}{y}{text}</code>	Same with filled node label
<code>\mdLabelRaw {x}{y}{text}</code>	Raw-text label at (x, y) .
<code>\mdLabelCenter {text}</code>	Math label at center (any aspect).
<code>\mdSpike {x}{y}{dir}</code>	L-corner; <code>dir</code> $\in \{\text{tl}, \text{tr}, \text{bl}, \text{br}\}$.
<code>\mdBlank</code>	Empty content (brackets only).
<code>\mdH</code>	Expands to the aspect value, i.e. the top edge's y -coordinate.

2 Main command

`\matrixdiagram[scale][aspect]{content}`

Inside `content`, the raw TikZ canvas runs $x \in [0, 1]$ and $y \in [0, \text{aspect}]$: `aspect` is literally the y -coordinate of the top edge, and `\mdH` expands to that value, so $(0, \text{\mdH})$ – $(1, \text{\mdH})$ is always the top edge, whatever the aspect. Every built-in positioning command (`\mdFill`, `\mdLabel`, `\mdLabelRaw`, `\mdHLine`, `\mdSpike`, and the ...From diagonal variants) instead takes y as a plain fraction of the height, $y \in [0, 1]$, and multiplies by `\mdH` internally — so `y=0.5` always means mid-height, regardless of aspect. Raw `\draw` commands written directly inside `content` see the unscaled canvas, so they must reference `\mdH` themselves, e.g. $(1, \text{\mdH})$ for the top-right corner.

2.1 Scale (aspect = 1)

`\matrixdiagram[1]{\mdBlank}` `\matrixdiagram[2]{\mdBlank}` `\matrixdiagram[3]{\mdBlank}`

$\left[\begin{array}{c} \\ \end{array} \right]$ $\left[\begin{array}{cc} & \\ & \end{array} \right]$ $\left[\begin{array}{ccc} & & \\ & & \end{array} \right]$

`\matrixdiagram[4]{\mdBlank}` `\matrixdiagram[5]{\mdBlank}`

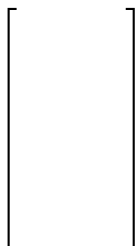
$\left[\begin{array}{ccc} & & \\ & & \end{array} \right]$ $\left[\begin{array}{cccc} & & & \\ & & & \end{array} \right]$

2.2 Aspect ratio (scale = 3)

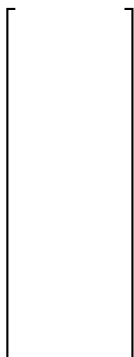
`\matrixdiagram[3][0.5]{\mdBlank}` `\matrixdiagram[3][1]{\mdBlank}` `\matrixdiagram[3][1.5]{\mdBlank}`

$\left[\begin{array}{cc} & \\ & \end{array} \right]$ $\left[\begin{array}{ccc} & & \\ & & \end{array} \right]$ $\left[\begin{array}{ccc} & & \\ & & \end{array} \right]$

```
$\matrixdiagram[3][2]{
  \mdBlank
}$
```



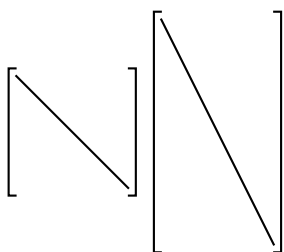
```
$\matrixdiagram[3][3]{
  \mdBlank
}$
```



3 Diagonal commands

3.1 `\mdDiag` — main diagonal

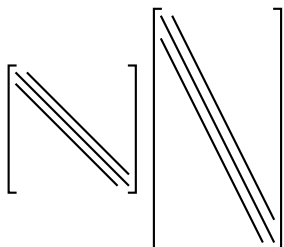
The single diagonal line from $(0, H)$ to $(1, 0)$.



```
$\matrixdiagram[3]{\mdDiag} \matrixdiagram[3][2]{\mdDiag}$
```

3.2 `\mdDiagBand` — tridiagonal band

Three parallel anti-diagonals with spacing $d = 0.1$.

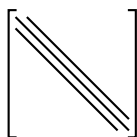


```
$\matrixdiagram[3]{\mdDiagBand} \matrixdiagram[3][2]{\mdDiagBand}$
```

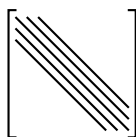
3.3 `\mdDiagBandN{n}` — variable band width

$n = 3, 5, 7, 9$ (spacing fixed at $d = 0.1$ per diagonal); `\mdDiagBandN{3}` is identical to `\mdDiagBand`:

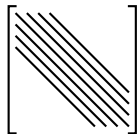
```
$\matrixdiagram[3]{\mdDiagBandN{3}}$
```



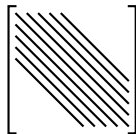
```
$\matrixdiagram[3]{\mdDiagBandN{5}}$
```



`\matrixdiagram[3]{\mdDiagBandN{7}}\$`

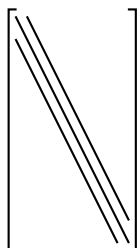


`\matrixdiagram[3]{\mdDiagBandN{9}}\$`

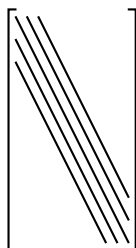


With `aspect = 2`:

`\matrixdiagram[3][2]{\mdDiagBandN{3}}\$`



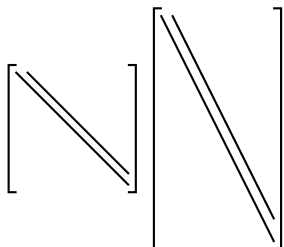
`\matrixdiagram[3][2]{\mdDiagBandN{5}}\$`



4 Bidiagonal commands

4.1 `\mdBidiagUpper` — main + super-diagonal

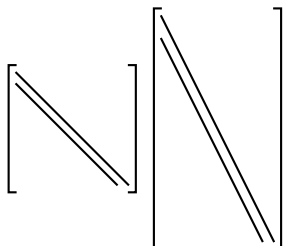
Main diagonal plus a single super-diagonal, offset $d = 0.1$ toward the top-right.



`\matrixdiagram[3]{\mdBidiagUpper}` `\matrixdiagram[3][2]{\mdBidiagUpper}\$`

4.2 `\mdBidiagLower` — main + sub-diagonal

Main diagonal plus a single sub-diagonal, offset $d = 0.1$ toward the bottom-left.

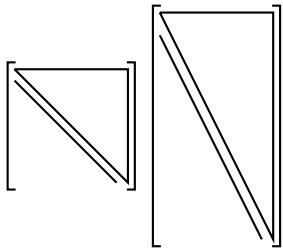


`\matrixdiagram[3]{\mdBidiagLower}` `\matrixdiagram[3][2]{\mdBidiagLower}\$`

5 Hessenberg command

5.1 `\mdHessenberg` — outline only (default)

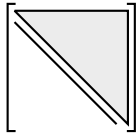
Upper-triangular outline $(0, H) \rightarrow (1, H) \rightarrow (1, 0)$ plus one sub-diagonal.



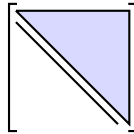
`\matrixdiagram[3]{\mdHessenberg}` `\matrixdiagram[3][2]{\mdHessenberg}`

5.2 `\mdHessenberg[color]` — with fill

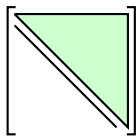
`\matrixdiagram[3]{\mdHessenberg[gray!15]}`



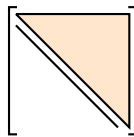
`\matrixdiagram[3]{\mdHessenberg[blue!15]}`



`\matrixdiagram[3]{\mdHessenberg[green!20]}`



`\matrixdiagram[3]{\mdHessenberg[orange!20]}`

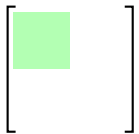


6 Fill and line commands

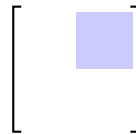
6.1 `\mdFill[color]{x0}{y0}{x1}{y1}`

The default color is gray!20:

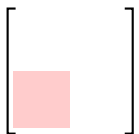
`\matrixdiagram[3]{`
`\mdFill[green!30]{0}{0.5}{0.5}{1}`
`}`



`\matrixdiagram[3]{`
`\mdFill[blue!20]{0.5}{0.5}{1}{1}`
`}`



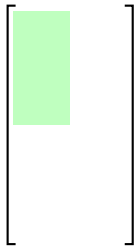
`\matrixdiagram[3]{`
`\mdFill[red!20]{0}{0}{0.5}{0.5}`
`}`

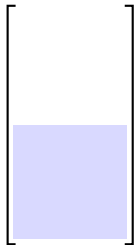


`\matrixdiagram[3]{`
`\mdFill{0}{0}{1}{1}`
`}`



With `aspect = 2` (y is still a fraction of the height, in $[0, 1]$):

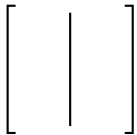


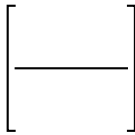
$$\text{\$}\text{\texttt{matrixdiagram}}[3][2]\{\text{\texttt{\mdFill}}[\text{green!25}]\{0\}\{0.5\}\{0.5\}\{1\}\}\text{\$}$$


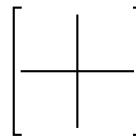
$$\text{\$}\text{\texttt{matrixdiagram}}[3][2]\{\text{\texttt{\mdFill}}[\text{blue!15}]\{0\}\{0\}\{1\}\{0.5\}\}\text{\$}$$

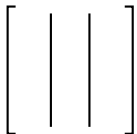
6.2 $\text{\texttt{\mdVLine}}\{x\}$ and $\text{\texttt{\mdHLine}}\{y\}$

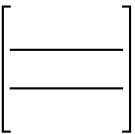
$\text{\texttt{\mdVLine}}\{x\}$ runs the full height at $x \in [0, 1]$; $\text{\texttt{\mdHLine}}\{y\}$ runs the full width at height $y \in [0, 1]$.

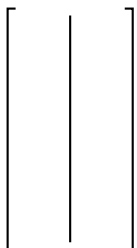
$$\text{\$}\text{\texttt{matrixdiagram}}[3]\{\text{\texttt{\mdVLine}}\{0.5\}\}\text{\$}$$


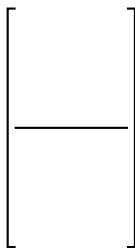
$$\text{\$}\text{\texttt{matrixdiagram}}[3]\{\text{\texttt{\mdHLine}}\{0.5\}\}\text{\$}$$


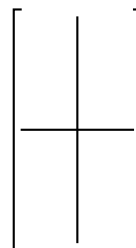
$$\text{\$}\text{\texttt{matrixdiagram}}[3]\{\text{\texttt{\mdVLine}}\{0.5\}\text{\texttt{\mdHLine}}\{0.5\}\}\text{\$}$$


$$\text{\$}\text{\texttt{matrixdiagram}}[3]\{\text{\texttt{\mdVLine}}\{0.33\}\text{\texttt{\mdVLine}}\{0.67\}\}\text{\$}$$


$$\text{\$}\text{\texttt{matrixdiagram}}[3]\{\text{\texttt{\mdHLine}}\{0.33\}\text{\texttt{\mdHLine}}\{0.67\}\}\text{\$}$$


$$\text{\$}\text{\texttt{matrixdiagram}}[3][2]\{\text{\texttt{\mdVLine}}\{0.5\}\}\text{\$}$$


$$\text{\$}\text{\texttt{matrixdiagram}}[3][2]\{\text{\texttt{\mdHLine}}\{0.5\}\}\text{\$}$$


$$\text{\$}\text{\texttt{matrixdiagram}}[3][2]\{\text{\texttt{\mdVLine}}\{0.5\}\text{\texttt{\mdHLine}}\{0.5\}\}\text{\$}$$


Combined with fills (block partitioning):

$$\left[\begin{array}{c|c} A & \\ \hline & D \end{array} \right]$$

```
$\matrixdiagram[3]{
  \mdFill[blue!15]{0}{0.5}{0.5}{1}
  \mdFill[green!15]{0.5}{0}{1}{0.5}
  \mdVLine{0.5}
  \mdHLine{0.5}
  \mdLabel{0.25}{0.75}{A}
  \mdLabel{0.75}{0.25}{D}
}
```

$$\left[\begin{array}{c|c} & b \\ \hline A & \end{array} \right]$$

```
$\matrixdiagram[3][2]{
  \mdFill[gray!15]{0}{0}{0.75}{1}
  \mdVLine{0.75}
  \mdLabel{0.375}{0.5}{A}
  \mdLabel{0.875}{0.5}{b}
}
```

7 Label commands

7.1 `\mdLabel{x}{y}{text}` — **math mode**

Places **text** in math mode at (x,y) ; the optional first argument fills the label's background node with that color.

```
$\matrixdiagram[3]{
  \mdLabel{0.2}{0.5}{A}
}
```

$$\left[\begin{array}{c} A \end{array} \right]$$

```
$\matrixdiagram[3]{
  \mdLabel[red!20]{0.5}{0.5}{Q^T}
}
```

$$\left[\begin{array}{c} Q^T \end{array} \right]$$

```
$\matrixdiagram[3]{
  \mdLabel{0.5}{0.5}{\Sigma}
}
```

$$\left[\begin{array}{c} \Sigma \end{array} \right]$$

```
$\matrixdiagram[3]{
  \mdDiagBand
  \mdLabel{0.25}{0.2}{T}
}
```

$$\left[\begin{array}{c} T \end{array} \right]$$

7.2 `\mdLabelCenter{text}` — **always centered**

Automatically placed at $(0.5, H/2)$ regardless of aspect ratio:

```

 $\matrixdiagram[3][1]{$ 
 $\mdLabelCenter{Q}$ 
 $\}$ 

```

$$\begin{bmatrix} Q \end{bmatrix}$$

```

 $\matrixdiagram[3][2]{$ 
 $\mdLabelCenter{Q}$ 
 $\}$ 

```

$$\begin{bmatrix} Q \end{bmatrix}$$

```

 $\matrixdiagram[3][3]{$ 
 $\mdLabelCenter{Q}$ 
 $\}$ 

```

$$\begin{bmatrix} Q \end{bmatrix}$$

```

 $\matrixdiagram[3][0.5]{$ 
 $\mdLabelCenter{Q}$ 
 $\}$ 

```

$$\begin{bmatrix} Q \end{bmatrix}$$

7.3 $\mdLabelRow{x}{y}{text}$ — raw text

Same placement as $\mdLabel$, but **text** is typeset as-is instead of in math mode — useful for plain words like “dense” or for text that already contains its own $\dots$.

$$\begin{bmatrix} \text{dense} \end{bmatrix} \begin{bmatrix} \approx 0 \end{bmatrix}$$

```

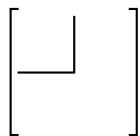
 $\matrixdiagram[3]{$ 
 $\mdLabelRow{0.5}{0.5}{dense}$ 
 $\}$ 
 $\matrixdiagram[3]{$ 
 $\mdLabelRow{0.5}{0.5}{\approx 0}$ 
 $\}$ 

```

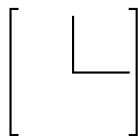
8 Spike commands

$\mdSpike{x}{y}{dir}$ draws an L-shaped corner at (x, y) . **tl** (top+left), **tr** (top+right), **bl** (bottom+left), **br** (bottom+right). Optionally, add a color fill via $\mdSpike[color]{x}{y}{dir}$.

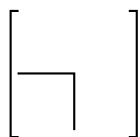

```
$\matrixdiagram[3]{
  \mdSpike{0.5}{0.5}{tl}
}$
```



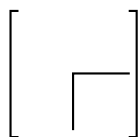
```
$\matrixdiagram[3]{
  \mdSpike{0.5}{0.5}{tr}
}$
```



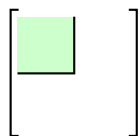
```
$\matrixdiagram[3]{
  \mdSpike{0.5}{0.5}{bl}
}$
```



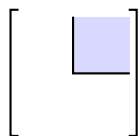
```
$\matrixdiagram[3]{
  \mdSpike{0.5}{0.5}{br}
}$
```



```
$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
}$
```

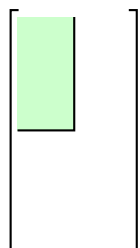


```
$\matrixdiagram[3]{
  \mdFill[blue!15]{0.5}{0.5}{1}{1}
  \mdSpike{0.5}{0.5}{tr}
}$
```

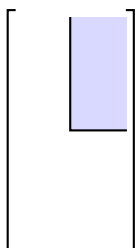


With aspect = 2, corner at mid-height (0.5, 0.5):

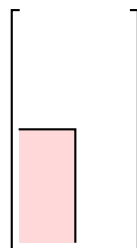
```
$\matrixdiagram[3][2]{
  \mdFill[green
    !20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
}$
```



```
$\matrixdiagram[3][2]{
  \mdFill[blue
    !15]{0.5}{0.5}{1}{1}
  \mdSpike{0.5}{0.5}{tr}
}$
```



```
$\matrixdiagram[3][2]{
  \mdFill[red
    !15]{0}{0}{0.5}{0.5}
  \mdSpike{0.5}{0.5}{bl}
}$
```

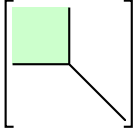


9 Clipped diagonal commands

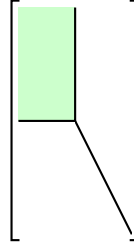
The ...From variants force the diagonal band from $[x_1, y_1]$ to $[x_2, y_2]$. Always pair with `\mdSpike` at the same corner.

9.1 `\mdDiagFrom{x1}{y1}{x2}{y2}`

```
$$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagFrom{0.5}{0.5}{1}{0}
}$
```

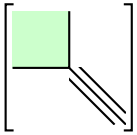


```
$$\matrixdiagram[3][2]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagFrom{0.5}{0.5}{1}{0}
}$
```

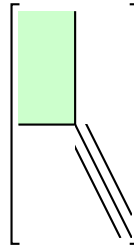


9.2 `\mdDiagBandFrom{x1}{y1}{x2}{y2}`

```
$$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandFrom{0.5}{0.5}{1}{0}
}$
```



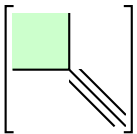
```
$$\matrixdiagram[3][2]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandFrom{0.5}{0.5}{1}{0}
}$
```



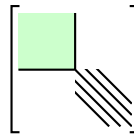
9.3 `\mdDiagBandNFrom{n}{x1}{y1}{x2}{y2}`

$n = 3, 5, 7$:

```
$$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandNFrom{3}{0.5}{0.5}{1}{0}
}$
```



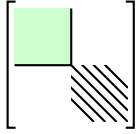
```
$$\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{tl}
  \mdDiagBandNFrom{5}{0.5}{0.5}{1}{0}
}$
```



```

 $\matrixdiagram[3]{
  \mdFill[green!20]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{t1}
  \mdDiagBandNFrom{7}{0.5}{0.5}{1}{0}
}$ 

```



10 Realistic equations

10.1 QR decomposition: $A = QR$

$$\begin{bmatrix} & \\ & \\ & \end{bmatrix} A = \begin{bmatrix} & \\ & \\ & \end{bmatrix} Q \begin{bmatrix} & \\ & \\ & \end{bmatrix} R$$

```

 $\matrixdiagram[3][2]{\mdLabelCenter{A}}$ 
=
 $\matrixdiagram[3][2]{\mdLabelCenter{Q}}$ 
 $\matrixdiagram[3]{\mdLabelCenter{R}}$ 

```

10.2 Bidiagonalization: $A = UBV^T$

$$\begin{bmatrix} & \\ & \\ & \end{bmatrix} A = \begin{bmatrix} & \\ & \\ & \end{bmatrix} U \begin{bmatrix} & \\ & \\ & \end{bmatrix} \begin{bmatrix} & \\ & \\ & \end{bmatrix} V^T$$

```

 $\matrixdiagram[3]{\mdLabelCenter{A}}$ 
=
 $\matrixdiagram[3]{\mdLabelCenter{U}}$ 
 $\matrixdiagram[3]{\mdBidiagUpper}$ 
 $\matrixdiagram[3]{\mdLabelCenter{V^T}}$ 

```

10.3 Similarity: $Q^T T_m Q = \tilde{T}_m$

$$\begin{bmatrix} & \\ & \\ & \end{bmatrix} Q^T \begin{bmatrix} & \\ & \\ & \end{bmatrix} T_m \begin{bmatrix} & \\ & \\ & \end{bmatrix} Q = \begin{bmatrix} & \\ & \\ & \end{bmatrix} \tilde{T}_m$$

```

 $\matrixdiagram[3]{\mdLabelCenter{Q^T}}$ 
 $\matrixdiagram[3]{
  \mdDiagBand
  \mdLabel{0.25}{0.15}{T_m}
}$ 
 $\matrixdiagram[3]{\mdLabelCenter{Q}}$ 
=
 $\matrixdiagram[3]{
  \mdFill[green!25]{0}{0.5}{0.5}{1}
  \mdSpike{0.5}{0.5}{t1}
  \mdDiagBandFrom{0.5}{0.5}{1}{0}
  \mdLabel{0.3}{0.12}{\widetilde{T}_m}
}$ 

```

10.4 Hessenberg reduction: $Q^T A Q = H$

$$\begin{bmatrix} Q^T \\ A \\ Q \end{bmatrix} = \begin{bmatrix} & & \\ & H & \\ & & \end{bmatrix}$$

```

 $\matrixdiagram[3]{\mdLabelCenter{Q^T}}
\matrixdiagram[3]{\mdLabelCenter{A}}
\matrixdiagram[3]{\mdLabelCenter{Q}}
=
\matrixdiagram[3]{
\mdHessenberg
\mdLabel{0.7}{0.55}{H}
}$ 

```

10.5 Schur complement structure

$$\begin{bmatrix} A & B \\ C & S \end{bmatrix}$$

```

 $\matrixdiagram[4]{
\mdFill[blue!10]{0}{0.5}{0.5}{1}
\mdFill[green!10]{0.5}{0}{1}{0.5}
\mdFill[gray!10]{0.5}{0.5}{1}{1}
\mdVLine{0.5}
\mdHLine{0.5}
\mdLabel{0.25}{0.75}{A}
\mdLabel{0.75}{0.75}{B}
\mdLabel{0.25}{0.25}{C}
\mdLabel{0.75}{0.25}{S}
}$ 

```